THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The centroid of the triangle with vertices $(-3, 2)$, $(3, 4)$, $(9, 0)$ has x-coordinate: 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

2. A circle has centre $(3, -1)$ and passes through $(12, 11)$. Its radius is: 1

(a) 14 **(b)** 17 **(c)** 16 **(d)** 15

3. The ratio $k : 1$ in which the line $x = 2$ divides the join of $(-2, 0)$ and $(6, 0)$, with $a < 2 < c$, has $k =$ 1

(a) 3 **(b)** 1 **(c)** 2 **(d)** 0

4. If $(1, 0)$, $(4, 6)$ and $(7, k)$ are collinear, then $k =$ 1

(a) 14 **(b)** 13 **(c)** 12 **(d)** 11

5. The perimeter of the triangle with vertices $(0, 0)$, $(7, 0)$, $(0, 24)$ is: 1

(a) 56 **(b)** 58 **(c)** 57 **(d)** 55

6. The point which divides $(10, 0)$ and $(0, 10)$ in the ratio 1:1 has coordinates summing to: 1

(a) 12 **(b)** 10 **(c)** 11 **(d)** 9

7. The centroid of the triangle with vertices $(4, 4)$, $(10, 2)$, $(1, 6)$ has x-coordinate: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 7

8. A circle has centre $(5, 5)$ and passes through $(8, 9)$. Its radius is: 1

(a) 5 **(b)** 4 **(c)** 7 **(d)** 6

9. The ratio $k : 1$ in which the line $x = 6$ divides the join of $(3, 0)$ and $(15, 0)$, with $a < 6 < c$, has $k =$ 1

(a) $\frac{1}{3}$ **(b)** $\frac{4}{3}$ **(c)** $\frac{7}{3}$ **(d)** $-\frac{2}{3}$

10. If $(-1, 1)$, $(1, 5)$ and $(3, k)$ are collinear, then $k =$ 1

(a) 11 **(b)** 10 **(c)** 9 **(d)** 8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the centroid of the triangle with vertices $(2,\ 2)$, $(8,\ 5)$ and $(5,\ 11)$. 2

12. Find the values of y for which the distance between $P(5,\ y)$ and $Q(0,\ 0)$ is 13 units. 2

13. Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 4)$, $B(5, 5)$ and $C(8, 9)$. Find the fourth vertex D. 2

14. Find the centroid of the triangle with vertices $(-4, 3)$, $(2, 9)$ and $(8, -3)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the points $(0, 1)$, $(6, 1)$ and $(3, 7)$ are the vertices of an isosceles triangle. 3

16. Find the coordinates of the point which divides the join of $(4, 1)$ and $(-8, 13)$ internally in the ratio $1 : 3$. 3

17. Find the area of the triangle with vertices $(-1, 1)$, $(5, 7)$ and $(3, -5)$. 3

OR

Show that the points $(2, -1)$, $(8, -1)$ and $(5, 4)$ are the vertices of an isosceles triangle. 3

18. Find the coordinates of the point which divides the join of $(-6, 4)$ and $(6, 10)$ internally in the ratio $5 : 1$. 3

OR

Find the area of the triangle with vertices $(1, 1)$, $(7, 3)$ and $(4, 8)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In what ratio does the x-axis divide the segment joining $(2, -4)$ and $(5, 8)$? Find the point of division. 5

OR

Find the point on the x-axis which is equidistant from $(0, 4)$ and $(6, 2)$. 5

20. In what ratio does the x-axis divide the segment joining $(-1, -7)$ and $(3, 5)$? Find the point of division. 5

OR

Find the point on the x-axis which is equidistant from $(3, 6)$ and $(11, 4)$. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND